

Rohan Jain

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SUMMARY

MS Data Science graduate from CU Boulder (GPA 3.9/4.0, May 2026) with production experience spanning AI/LLM systems, backend software engineering, and applied machine learning - most recently as an AI Engineering Intern at Bright Machines building production LLM agent infrastructure, and previously at Nexus Weather and Climate building end-to-end ML pipelines. Comfortable moving between model development, backend systems, and data infrastructure, with a consistent focus on reliability. Open to Software Engineer, Data Scientist, ML Engineer, AI Engineer, and Data Engineer roles.

SKILLS

AI & Machine Learning: LangGraph, LangChain, Model Context Protocol (MCP), LiteLLM, RAG, embeddings, BM25, OpenAI / Anthropic / Gemini APIs, LLM evaluation harnesses; TensorFlow, PyTorch, scikit-learn, XGBoost, Random Forest, MLflow

Backend & Infrastructure: Python, FastAPI, REST APIs, Pydantic v2, PostgreSQL, Alembic, SQLite / sqllite-vec, Docker, Kubernetes, Kustomize, ArgoCD, Temporal, pytest, GitLab CI/CD, GitHub Actions

Cloud, Data & Frontend: Azure (AKS, ACR, Entra ID), AWS (Lambda, S3, EC2, API Gateway, RDS), Databricks (Unity Catalog), PostGIS, ETL pipeline design; TypeScript, React, Next.js; Grafana, Loki, Prometheus, Langfuse

Languages & Tools: Python, R, SQL (PostgreSQL, MySQL), TypeScript, Bash; Git/GitHub, Tableau, Power BI

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Owned tool selection for an on-prem LLM agent (**LangGraph** ReAct over **LiteLLM**): 132 **MCP** tool schemas exceeded the 32K-token window; built a 115-question harness that raised recall 0.47 to 0.97.
- Designed and shipped a read-only-by-construction **MCP** server from scratch (37 files, +3,163 lines, 12 tools) serving a 32,000-workflow **Temporal** namespace in production; Docker-packaged, deployed via **Kubernetes/Kustomize**.
- Hardened four **MCP** servers behind a production LLM agent: closed a Critical path-traversal vulnerability (**CWE-22**) with 9 regression tests, removed cluster-secret access from a **Kubernetes** service account, enforced least-privilege RBAC.
- Root-caused a **74-minute** silent chat outage (agent registry not rehydrating after a gateway restart while health probes stayed green), disproved the planned upgrade fix by inspecting the running container, and shipped **Postgres** persistence plus a self-healing watchdog (**135 tests**), verified live in production.
- Led the agent-orchestration architecture spike for a production **LangGraph** agent: evaluated the **A2A** (agent-to-agent) registry protocol against Temporal and **DBOS** as a durable backbone, showed transport is not durability, and published a stick/adopt/hybrid roadmap to Confluence for team review.

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Designed a probabilistic CNN-LSTM with a custom CRPS loss for calibrated ensemble uncertainty quantification, packaging reproducible artifact bundles and achieving a best-reported CRPS of 0.1787.
- Implemented an automated inference quality gate rejecting models whose post-ML MAE fails to beat the pre-ML baseline; reduced failed production runs 90% and saved 20 engineer-hours monthly.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Reverse-engineered proprietary Thermo Isodat binary formats via DLL decompilation, building deterministic R readers; migrated lab infrastructure to AWS EC2 with Keycloak OIDC and Grafana/Loki/Prometheus observability.

EDUCATION

University of Colorado, Boulder

Aug 2024 – May 2026 | Boulder, CO

Master's in Data Science (GPA: 3.9/4.0)

Mumbai University, Rajiv Gandhi Institute of Technology

Jul 2020 – May 2024 | Mumbai, India

B.E. in Computer Engineering (GPA: 3.7/4.0)

PROJECTS

ClaimPilot AI - Guardrailed LLM Agent | Python, Pydantic v2, OpenAI function calling

Jan 2026

- Architected a hybrid agent for healthcare claims validation: a deterministic Pydantic layer validates codes while OpenAI function calling generates structured fixes; strict schemas with retry logic guarantee a valid report.

AgentSquared - No-Code AI Agent Builder | Python, FastAPI, SQLAlchemy, Next.js, Tailwind, Gemini API

Mar 2026

- Built a no-code AI agent platform at HackCU 12: config-driven architecture where every agent is a database row plus JSON config, Gemini-powered RAG, and real-time Bluesky brand monitoring with human-approved replies.

PUBLICATION

- Yoga Posture Detection and Correction - published research paper, 96.5% accuracy
(journals.stmjournals.com/joosdt/article=2024/view=161704/)